

CSE:4227 Digital Image Processing (Sec- A, B, C)

Fall 2022 (2 May 2023 – 24 August 2023)

Class-wise Topics/Contents: (Mon/Wed/Thurs:: each class 50 mint.)

Instructor: Prof. Dr. Kazi A Kalpoma and [Sanzana Karim Lora](#)

<u>Week/Class</u>		<u>Date</u>	<u>Topics covered</u>
<u>1</u>	C_1	03/05	Instructor's Introduction , Course policy, Course outline, Calendar with class-wise topics distribution. Introduction to the DIP course.
	C_2	04/05	GOV. Holiday
<u>2</u>	C_3	08/05	Chap#1: Introduction: What is Digital Image? Image Formation and Image Digitization. What is Digital Image Processing (DIP)? Why do we need DIP? Key stages in DIP and how they work.
	C_4	10/05	Chap#2: Digital Image Fundamentals: Visual Perception and Image Formation in an Eye, Image Sensing, and Acquisition , Representation of Digital Image.
	C_5	11/05	Chap#2: Digital Image Fundamentals: Spatial and Intensity Resolution, Effect of Spatial and Gray Level Resolutions, Image Scaling (Zooming and Shrinking) process, Nearest neighbor interpolation.
<u>3</u>	C_6	15/05	Chap#2: Digital Image Fundamentals: Some basic relationships between Pixels - neighborhood, adjacency, connectivity, paths, and regions. Distance measures (City block, Chessboard, and Euclidean distance).
	C_7	17/05	Revision and Solving Exercises on Chap#2
	C_8	18/05	Chap#3 Intensity Transformations & Spatial Filtering: Some basic Intensity Transformation functions – Linear, Log, Power Law (Gamma) for Image Enhancement.

<u>4</u>	C_9	22/05	Chap#3 Intensity Transformations & Spatial Filtering (cont.): Piecewise Linear Transformation (Contrast Stretching, Intensity Level Slicing, Bit Plane Slicing).
	C_10	24/05	Revision on Intensity Transformation Functions
	C_11	25/05	Quiz-1
<u>5</u>	C_12	29/05	Chap#3 Intensity Transformations & Spatial Filtering (cont.): Image Histogram Processing - Image histogram, Use of histogram, Normalized histogram, Histogram processing techniques: histogram Sliding, Stretching histogram.
	C_13	31/05	Chap#3 Intensity Transformations & Spatial Filtering (cont.): Image Histogram Equalization - Need of image histogram equalization, histogram equalization algorithm, equalization transform functions.
	C_14	01/06	Chap#3 Intensity Transformations & Spatial Filtering (cont.): Histogram Matching (specification), Revision on histogram
<u>6</u>	C_15	05/06	Chap#3 Intensity Transformations & Spatial Filtering (cont.): Fundamentals of Spatial Filtering - What is masking? Correlation & Convolution operations, Smoothing spatial filters - (mean, weighted mean filter).
	C_16	07/06	Chap#3 Intensity Transformations & Spatial Filtering (cont.): Non-linear Min, Max, and Median filter, Different filtering designs and their applications.
	C_17	08/06	Chap#3 Intensity Transformations & Spatial Filtering (cont.): Sharpening spatial Filters - Image derivatives, Laplacian Filter, Laplacian Image Enhancement, Difference filters, Combining filtering techniques.
<u>7</u>	C_18	12/06	Quiz-2

	C_19	14/06	Chap#9 Morphological Image Processing (Part A): What is morphology? Basic Set Theory, Structuring Element, Basic operations: Erosion, Dilation, Applications of basic morphological operations
	C_20	15/06	Opening, Closing Morphological Image Processing, Compound operations. Morphological Algorithms: Boundary extraction. Region filling
<u>8</u>	C_21	19/06	Chap#9 Morphological Image Processing (Part B): Shape or Boundary Detection by Hit and Miss Transform. Morphological Algorithms.
	C_22	21/06	Chap#9 Morphological Image Processing (Part B): Thinning and Thickening Morphological Process
	C_23	22/06	Revision and Solving Exercises on Chap#9
<u>9</u>		23/06	EID + MID BREAK
		06/07	
<u>10</u>	C_24	10/07	Chap#10 Image Segmentation: Definition, segmentation problems, different segmentation techniques, Introduction of edges, Finding points and line detection.
	C_25	12/07	Chap#10 Image Segmentation: Edge detection approach for segmentation. Canny edge detection and edge linking.
	C_26	13/07	Chap#10 Image Segmentation: Line detection using Hough Transform.
<u>11</u>	C_27	17/07	Chap#10 Image Segmentation: Thresholding and Region-Based Segmentation.

	C_28	19/07	Revision and Solving Exercises on Chap#10
	C_29	20/07	Quiz-3
12	C_30	24/07	Chap#6 Color Image Processing: Colour fundamentals and Color models.
	C_31	26/07	Chap#6 Color Image Processing: RGB and HSI Color Model.
	C_32	27/07	Revision and solving exercises on Chap#6
13	C_33	31/07	Chap#8 Image Compression: Why do we need compression? Compression Ratio and Relative Data Redundancy. How can we implement compression: Coding redundancy, Spatial and temporal redundancy, and Irrelevant information.
	C_34	02/08	Chap#8 Image Compression: Information theory review, Image Entropy, Lossy and Loss-free Compression, Different Compression Techniques: Huffman coding.
	C_35	03/08	Chap#5 Image Restoration: What is image restoration? Noise models, Noise removal using spatial domain filtering.
14	C_36	07/08	Chap#8 Image Compression: Different Compression Techniques: Golomb coding, Arithmetic coding.
	C_37	09/08	Chap#8 Image Compression: Different Compression Techniques: LZW coding, Run-length encoding (RLE), RLE on Binary Image, RLE on BMP.
	C_38	10/08	Chap#11 Image Representation and Description: Chain Code and Image Fidelity Criteria.
15	C_39	14/08	Revision and solving exercises on Chap#8

	C_40	16/08	Quiz-4
	C_41	17/08	Chap#11 Spatial Feature Extraction: What is feature extraction in DIP? Where is feature extraction used? What is the advantage of feature extraction?
<u>16</u>	C_42	21/08	Chap#11 Spatial Feature Extraction: Different types of feature extractions.
	C_43	23/08	Revision Class
	C_44	24/08	Farewell